16/01/25

Tutorial -1

Q1) (a) Considering classical electrodynamics, compute the peak electric field generated by a 5 watt LED bulb and passing through a square of side 10 cm?

Sol) • energy flux 'S' coming out from the bulb & hitting a square area of side 10 cm is given by,

$$\Rightarrow S = \frac{5 \text{ Watt}}{100 \text{ cm}^2} = \frac{5 \times 100 \times 100}{100} \text{ Js}^{-1}\text{m}^{-2} = 500 \text{ Jm}^{-2}\text{s}^{-1}$$

$$\therefore \boxed{S = 500 \text{ Jm}^{-2}\text{s}^{-1}}$$

• The energy flux of a classical EM wave is given by the 'average of the magnitude of poynting vector' given by

$$\Rightarrow \langle S \rangle = \frac{1}{2} \varepsilon_0 c E_0^2$$

$E_0 \rightarrow$ amplitude/peak of the electric field.
$\varepsilon_0 \rightarrow$ dielectric constant in vacuum
$c \rightarrow$ speed of light in vacuum

$$\Rightarrow E_0 = \left(\frac{2\langle S \rangle}{\varepsilon_0 c} \right)^{1/2} \simeq 612 \text{ Volt/m}$$

$$\therefore \boxed{E_0 = 612 \text{ Volts/m}}$$

(b) if this LED bulb emits red light with $\lambda = 650$ nm, then as per Planck's proposal, how many light quanta pass through the square per second.

Sol) Planck's proposal,

$$\Rightarrow E = nh\nu$$

h = planck's const.
ν = frequency of photon
n = no. of photons
E = Total energy of photons

$$\Rightarrow S = n'h\nu$$

S = Total energy per unit area per unit time of photons
n' = no. of photons passing per unit area per unit time.

$$\Rightarrow n' = \frac{S}{h\nu} = 1.6 \times 10^{21}$$

$\therefore$ no. of photons passing through the square of $10 \times 10 \text{ cm}^2 = 1.6 \times 10^{19} \text{ Js}^{-1}$

Q2) An experimentalist determined the wavelength of an electromagnetic radiation from hydrogen gas to be 1010 nm using a spectrometer having a maximum error of 1%. Using Bohr model, determine the initial and final energy level of the electron that caused the aforementioned radiation. given the ground state energy of hydrogen atom, $E_1 = -13.6$ eV.

Sol)

→ The quantised energy levels of Bohr electron is given by,

$$\Rightarrow E_n = -\frac{13.6}{n^2} \text{ eV}$$

→ energy of an emitted photon is given by,

$$\Rightarrow E = \frac{hc}{\lambda} = 1.2 \text{ eV} \quad \text{– for } \lambda = 1010 \text{ nm}$$

⇒ Since λ is actually 1010 ± 10.1 nm

$$\therefore \boxed{E = 1.2 \pm 0.02 \text{ eV}}$$

→ now, we have to find two levels whose energy gap falls in the interval $(1.22, 1.18)$ eV.

→ it will come out to be $7 \rightarrow 3$

Q3) Consider the first 25 digits in the decimal expansion of π (3, 1, 4, 1, 5, 9, 2, 6, 5, 3, ...) → events (x)

(a) if you selected one number at random, from this set, what are the probabilities of getting each of the 10 digits?

Sol) $$P(x_i) = \frac{\text{no. of events where } x_i \text{ happened}}{\text{Total no. of events that happened}}$$

$x_i = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9$

(b) what is the most probable digit? what is the median digit? what is the average value?

Sol) ① most probable digit = $\max(P(x_1), P(x_2) \ldots P(x_{10}))$

② median digit = center value of the set

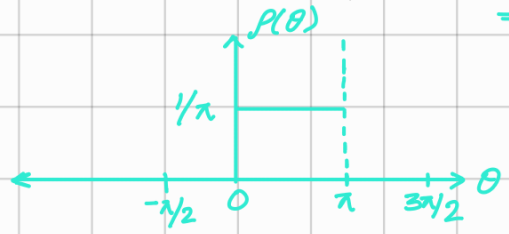
③ average value = $\frac{1}{25}\sum_{i=1}^{10} f_i x_i$, f_i = frequency of x_i

④ Standard deviation ⇒ $\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2 = \left(\frac{1}{25}\sum_{i=1}^{10} f_i x_i^2\right) - \left(\frac{1}{25}\sum_{i=1}^{10} f_i x_i\right)^2$

Q) The needle on a broken car speedometer is free to swing, and bounces perfectly off the pins at either end, so that if you give it a flick it is equally likely to come to rest at any angle between '0' and 'π'.

(a) What is the probability density, $\rho(\theta)$?

Sol)



⇒ for a random variable x with pdf $p(x)$: ① $\int_{-\infty}^{\infty} p(x)\,dx = 1$

② $\langle g(x)\rangle = \int_{-\infty}^{\infty} g(x)\,p(x)\,dx$

$$\boxed{\Rightarrow \int_0^{\pi} \rho(\theta)\,d\theta = 1 \quad \therefore \rho(\theta) = 1/\pi}$$

(b) Compute $\langle\theta\rangle$, $\langle\theta^2\rangle$ & σ.

Sol)

$$\boxed{\langle\theta\rangle = \int_{-\infty}^{\infty} \theta\,\rho(\theta)\,d\theta = \pi/2}$$

$$\boxed{\langle\theta^2\rangle = \int_{-\infty}^{\infty} \theta^2\,\rho(\theta)\,d\theta = \pi^2/3}$$

$$\boxed{\sigma = \sqrt{\langle\theta^2\rangle - \langle\theta\rangle^2} = \pi/2\sqrt{3}}$$

(c) Compute $\langle\sin\theta\rangle$, $\langle\cos\theta\rangle$ & $\langle\cos^2\theta\rangle$

Sol)

$$\boxed{\langle\sin\theta\rangle = \int_{-\infty}^{\infty} \sin\theta\,\rho(\theta)\,d\theta = 2/\pi}$$

$$\boxed{\langle\cos\theta\rangle = \int_{-\infty}^{\infty} \cos\theta\,\rho(\theta)\,d\theta = 0}$$

$$\boxed{\langle\cos^2\theta\rangle = \int_{-\infty}^{\infty} \cos^2\theta\,\rho(\theta)\,d\theta = 1/2}$$